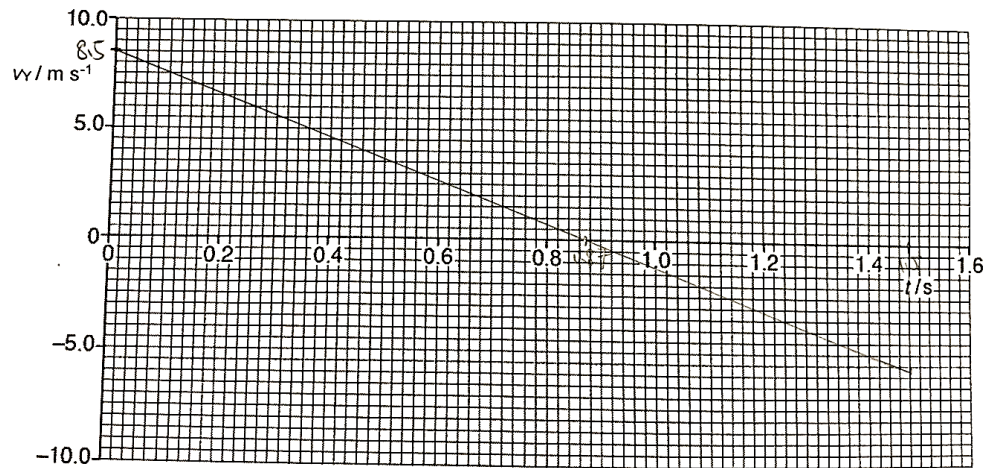


Section A

- 1 (a) change in velocity / time (taken) or rate of change of velocity **B1**
- (b) (i) $v_x = (24 / 1.5) = 16 \text{ m s}^{-1}$ **A1**
- (ii) $\tan 28^\circ = v_y / v_x$ or $v_x = v \cos 28^\circ$ and $v_y = v \sin 28^\circ$ **M1**
 $v_y = 16 \tan 28^\circ$ or $v_y = 16 \times (\sin 28^\circ / \cos 28^\circ)$ so $v_y = 8.5 \text{ m s}^{-1}$ **A0**
- (iii) $v = u + at$ **C1**
 $t = (0 - 8.5) / (-9.81)$
 $= 0.87 \text{ s}$ **A1**
- (iv) straight line from positive v_y at $t = 0$ to negative v_y at $t = 1.5 \text{ s}$ **M1**
line starts at $(0, 8.5)$ and crosses t -axis at $(0.87, 0)$ and does not go beyond $t = 1.5 \text{ s}$. **A1**



- (c) $v^2 = u^2 + 2as$
 $0 = 8.5^2 + 2(-9.81)s$
or $(s = ut + \frac{1}{2}at^2)$ $s = 8.5 \times 0.87 + \frac{1}{2} \times (-9.81) \times 0.87^2$
or $(s = vt - \frac{1}{2}at^2)$ $s = 0 - \frac{1}{2} \times (-9.81) \times 0.87^2$
or $(s = \frac{1}{2}(u + v)t \text{ or area under graph})$ $s = 0.5 \times 8.5 \times 0.87$ **C1**
 $s = 3.7 \text{ m}$ **A1**

2

- (d)** acceleration (of freefall) is unchanged / not dependent on mass, and so no effect (on maximum height)

OR explanation in terms of energy:

(initial) $KE \propto \text{mass}$, $(\Delta)KE = (\Delta)PE$, (max) $PE \propto \text{mass}$, and so
no effect (on maximum height)

B1

2

- (a)** Force experienced by a mass in a gravitational field

B1